

Ranked Nutrition Interventions for LMIC Government Scaling

Source file: /Users/akashshaji/Documents/GitHub/nutri-evidence-review/top_papers_for_review.json

Reviewed set: 100 ranked papers from the provided database. The file contains 57 records with structured PMC full text and 43 abstract-only records. Most records are PubMed-indexed meta-analyses or systematic reviews; the only record marked `nutrition-sensitive` in the top 100 is an OpenAlex/Lancet migration review, while nutrition-sensitive agriculture/cash-transfer evidence appears mainly in supplementary PubMed reviews.

Important extraction rule: effect sizes below are reported only when present in the provided abstract or PMC-derived text/tables. Where the record is abstract-only, the effect size is flagged as abstract-only. Cost-effectiveness ratings are implementation judgments informed by the database and general nutrition-program economics; the provided top 100 did not contain a distinct `CEA` tier record.

Intervention Universe

Nutrition-specific interventions identified in the 100 papers:

1. Vitamin A supplementation for children 6-59 months.
2. Neonatal/early-infancy vitamin A supplementation.
3. Vitamin A/carotenoid supplementation in pregnancy or postpartum.
4. Iron-folic acid supplementation in pregnancy, including intermittent regimens.
5. Nutrition education/counseling to improve iron-folic acid adherence.
6. Maternal folic acid supplementation.
7. Maternal multiple micronutrient supplementation (MMS/MMN).
8. Maternal vitamin B12, vitamin D, calcium, zinc, iodine and other single-micronutrient supplementation.
9. Multiple micronutrient supplementation for lactating women.
10. Preventive zinc supplementation for children.
11. Therapeutic zinc for childhood diarrhea.
12. Large-scale staple food fortification with iron, iodine, folic acid or vitamin A.
13. Double-fortified salt with iron and iodine.
14. Micronutrient powders (MNPs) for home fortification.
15. Fortified complementary foods and fortified non-dairy beverages.
16. Complementary feeding education/counseling.
17. Complementary food provision or supplementary feeding.
18. Small-quantity lipid-based nutrient supplements (SQ-LNS) for children or pregnancy.
19. Ready-to-use therapeutic/supplementary foods for SAM/MAM, including RUTF/RUSF and fortified blended foods.
20. Balanced protein-energy supplementation in pregnancy.
21. Breastfeeding promotion, support, and exclusive breastfeeding protection.
22. Nutritional management of low-birthweight/preterm or small nutritionally at-risk infants.
23. School-age/adolescent iron, iron-folic acid, and MMN supplementation/fortification.
24. Egg-based supplementation.
25. Nutritional support for adults living with HIV.
26. Nutrition education/mHealth interventions in pregnancy.
27. Growth monitoring and promotion as a delivery platform.
28. Integrated child-health/nutrition packages, including IMCI/IMNCI.

Nutrition-sensitive interventions identified:

- 29. Financial incentives, vouchers, user-fee removal, and conditional cash transfers linked to child health/nutrition service use.
- 30. Agriculture/home-garden/biofortification/livestock and food-system interventions.
- 31. WASH interventions linked to nutrition.
- 32. Agricultural and nutrition education to reduce aflatoxin exposure.

Context-only or not ranked as interventions: undernutrition/anemia/rickets prevalence reviews, dietary diversity determinant reviews, cryptosporidium risk-factor reviews, parental migration as an exposure, early postnatal discharge, pneumonia case management, and surgical oral nutrition support.

Summary Table

Rank	Intervention	Evidence	Cost-effectiveness	Scalability	Target
1	Vitamin A supplementation, 6-59 mo	A	Very High	Proven national	Children 6-59 mo in deficient/high-risk settings
2	Large-scale food fortification	A	Very High	Proven national	General population, women, children
3	Iron-folic acid in pregnancy plus adherence support	A	Very High	Proven national	Pregnant women
4	Maternal MMS replacing IFA where feasible	A	Very High	Proven national platform / growing policy adoption	Pregnant women
5	Therapeutic zinc with ORS for diarrhea	A	Very High	Proven national	Children older than 6 mo with diarrhea
6	Breastfeeding promotion and protection	A	Very High	Proven national	Infants 0-6 mo and mothers
7	Micronutrient powders for home fortification	A	High	Proven subnational / national in many settings	Children 6-23 mo

Rank	Intervention	Evidence	Cost-effectiveness	Scalability	Target
8	Complementary feeding counseling plus food where needed	A	High	Proven subnational	Children 6-23 mo
9	CMAM/RUTF/RUSF for acute malnutrition	A	High	Proven subnational / national in emergencies	Children with SAM/MAM
10	SQ-LNS for prevention	A	Moderate-High	Growing	Children 6-23 mo; pregnant women
11	Fortified complementary foods/beverages	B+	Moderate-High	Growing	Infants/young children and school-age children
12	Double-fortified salt	B+	Very High	Proven subnational / national potential	General population
13	Balanced protein-energy in pregnancy	B+	Moderate	Requires targeting investment	Undernourished pregnant women
14	School-age/adolescent iron/MMN/WIFAS	B+	High	Proven subnational	School-age children, adolescents
15	Preventive zinc supplementation	B+	High	Proven subnational	Children 6 mo-12 y
16	Financial incentives/cash-plus nutrition	B	Moderate	Proven national social-protection platform	Poor households with children
17	WASH integrated with nutrition	B	Moderate	Requires infrastructure investment	Children under 5 and households
18	Nutrition-sensitive agriculture/eggs/food systems	B-/C+	Moderate	Growing	Food-insecure households
19	Growth monitoring and promotion	C+	Moderate as a platform	Proven national	Children under 5

	Rank	Intervention	Evidence	Cost-effectiveness	Scalability	Target
	20	LBW/preterm infant nutritional management	C+	Moderate	Growing	LBW/preterm infants
	21	HIV adult nutritional support	C+	Unknown/Moderate	Requires investment	Adults living with HIV
	22	Aflatoxin-reduction education/agriculture	C	Unknown	Requires investment	Infants/children in aflatoxin-prone areas

Tier 1: Strong Evidence, Proven Scale, High Value

1. Vitamin A supplementation for children 6-59 months

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national. Nutrition-specific.

Evidence justification: This is one of the strongest clusters in the database, validated by PubMed publication types including Meta-Analysis and Systematic Review and MeSH terms around vitamin A, child mortality, developing countries, and randomized trials. The key full-text paper is Imdad et al. 2011, BMJ, PMID 21868478, 349 citations, pmc . It included 43 randomized trials and reports: all-cause mortality rate ratio 0.76 (95% CI 0.69 to 0.83; 17 trials, 194,483 participants); diarrhea mortality 0.72 (95% CI 0.57 to 0.91); diarrhea incidence 0.85 (95% CI 0.82 to 0.87); measles incidence 0.50 (95% CI 0.37 to 0.67); night blindness 0.32 (95% CI 0.21 to 0.50); vomiting within 48 hours 2.75 (95% CI 1.81 to 4.19).

The more recent Cochrane update is abstract-only in the database: PMID 35294044, 2022, Cochrane Database, 72 citations, reports all-cause mortality RR 0.88 (95% CI 0.83 to 0.93) and diarrhea mortality RR 0.88 (95% CI 0.79 to 0.98). The 2017 Cochrane abstract-only record, PMID 28282701, 487 citations, reports the same fixed-effect mortality estimate and notes a random-effects estimate of RR 0.76 (95% CI 0.66 to 0.88).

Subgroups/modifiers: The database supports routine child VAS in settings with vitamin A deficiency or high child mortality. It also includes a Nepal-focused full-text synthesis, PMID 35584136, 2022, 2 citations, suggesting diminishing absolute returns where deficiency and mortality have fallen. Neonatal and infant VAS records (PMID 28234402 and PMID 21975770) are abstract-only and more mixed, so they are not grouped with the Tier 1 child 6-59 month program.

Mechanism: Retinoids maintain epithelial integrity, visual-cycle function, immune response, and resistance to severe diarrhea/measles outcomes.

Government scaling pathway: Semiannual child health days, immunization contacts, community health-worker distribution, and stock monitoring. This is a classic high-coverage campaign intervention, but targeting should tighten as deficiency prevalence declines.

Key supporting papers: PMID 21868478 (2011, BMJ, 349 cites, PMC); PMID 35294044 (2022, Cochrane, 72 cites, abstract-only); PMID 28282701 (2017, Cochrane, 487 cites, abstract-only); PMID 21501438 (2011, BMC Public Health, 155 cites, PMC).

2. Large-scale food fortification

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national. Nutrition-specific, population-wide.

Evidence justification: Keats et al. 2019, American Journal of Clinical Nutrition, PMID 30997493, 215 citations, [pmc](#), is the main full-text source. It covers 136 LSFF studies and 50 meta-analyzed studies from LMICs. Full-text tables report vitamin A fortification increased serum retinol, SMD 0.31 (95% CI 0.18 to 0.45); salt iodization increased urinary iodine, SMD 1.02 (95% CI 0.63 to 1.42), reduced iodine deficiency, RR 0.25 (95% CI 0.21 to 0.29), and reduced goiter, OR 0.26 (95% CI 0.16 to 0.43). Iron fortification improved hemoglobin, SMD 0.19 (95% CI 0.04 to 0.35), reduced anemia, RR 0.66 (95% CI 0.59 to 0.74), improved ferritin, SMD 0.39 (95% CI 0.34 to 0.44), and reduced iron deficiency, RR 0.42 (95% CI 0.32 to 0.56). Folic acid fortification reduced total neural tube defects, OR 0.59 (95% CI 0.49 to 0.70), spina bifida OR 0.66 (95% CI 0.53 to 0.82), and anencephaly OR 0.49 (95% CI 0.40 to 0.60).

Double-fortified salt is a related but distinct vehicle. The database has it as abstract-only: PMID 29767699, 2018, Advances in Nutrition, 38 citations. It reports efficacy-study hemoglobin SMD 0.28 (95% CI 0.11 to 0.44), anemia RR 0.59 (95% CI 0.46 to 0.77), and iron-deficiency anemia RR 0.37 (95% CI 0.25 to 0.54). Because DFS effectiveness data were smaller (hemoglobin SMD 0.03, 95% CI 0.01 to 0.05), DFS is ranked separately in Tier 2 even though the cost profile is excellent.

Mechanism: Adds limiting micronutrients to centrally processed staples or condiments, increasing intake without requiring daily adherence to pills or sachets.

Government scaling pathway: Mandatory standards for flour, rice, oil, sugar or salt; food-control inspection; premix procurement; industry compliance incentives; biomarker surveillance. Best suited where a large share of staples pass through inspectable mills/refineries.

Key supporting papers: PMID 30997493 (2019, AJCN, 215 cites, PMC); PMID 31849042 (2019, Cochrane, 87 cites, abstract-only); PMID 29767699 (2018, Advances in Nutrition, 38 cites, abstract-only); PMID 30880739 (2019, J Health Popul Nutr, 8 cites, abstract-only).

3. Iron-folic acid supplementation in pregnancy plus adherence support

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national. Nutrition-specific.

Evidence justification: IFA is an established ANC intervention, with implementation papers in the database highlighting that adherence is the binding constraint. In a full-text meta-analysis of nutrition education during pregnancy, PMID 39545365, 2025, Nutrition Reviews, 17 citations, [pmc](#), education improved IFAS compliance, OR 2.80 (95% CI 2.04 to 3.83), increased hemoglobin, MD 0.88 g/dL (95% CI 0.43 to 1.34), and reduced anemia, OR 0.58 (95% CI 0.35 to 0.97). A full-text sub-Saharan Africa compliance review, PMID 33852614, 2021, PLoS One, 66 citations, [pmc](#), found pooled compliance of 39.2%; counseling was associated with higher adherence, OR 1.96 (95% CI 1.76 to 5.93), knowledge of IFAS OR 2.71 (95% CI 1.33 to 5.54), and knowledge of anemia OR 5.42 (95% CI 1.52 to 19.43).

On dosing, the abstract-only Cochrane record PMID 26482110, 2015, 256 citations, found intermittent regimens showed no clear infant-outcome difference compared with daily regimens: LBW RR 0.82 (95% CI 0.55 to 1.22), birthweight MD 5.13 g (95% CI -29.46 to 39.72), and fewer side effects RR 0.56 (95% CI 0.37 to 0.84). A full-text 2025 review, PMID 39780191, Reproductive Health, 3 citations, similarly found intermittent regimens had lower hemoglobin than daily regimens, MD -0.24 g/dL (95% CI -0.35 to -0.12), but lower gastric side effects RR 0.27 (95% CI 0.11 to 0.69), better adherence RR 1.60 (95% CI 1.34 to 1.91), and no clear anemia-incidence difference RR 1.09 (95% CI 0.77 to 1.54).

Maternal folic acid evidence is also strong: PMID 31680411, 2020, Maternal & Child Nutrition, 30 citations, [pmc](#), reports mean birthweight MD 0.37 kg (95% CI 0.24 to 0.50) and LBW OR 0.59 (95% CI 0.47 to 0.74).

Mechanism: Iron supports hemoglobin and oxygen transport; folate supports DNA synthesis and neural-tube closure. Counseling addresses forgetfulness, side effects, and risk perception.

Government scaling pathway: ANC tablet distribution, counseling scripts, community follow-up, side-effect management, commodity tracking, and mHealth reminders where feasible.

Key supporting papers: PMID 39545365 (2025, Nutrition Reviews, 17 cites, PMC); PMID 33852614 (2021, PLoS One, 66 cites, PMC); PMID 26482110 (2015, Cochrane, 256 cites, abstract-only); PMID 39780191 (2025, Reproductive Health, 3 cites, PMC); PMID 31680411 (2020, Maternal & Child Nutrition, 30 cites, PMC).

4. Maternal multiple micronutrient supplementation

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national platform, growing product transition. Nutrition-specific.

Evidence justification: The maternal MMS cluster has several meta-analyses and consistent birth-size benefits. The abstract-only 2019 Cochrane update, PMID 30873598, Cochrane Database, 364 citations, reports MMS with iron-folic acid reduced LBW, RR 0.88 (95% CI 0.85 to 0.91), and SGA, RR 0.92 (95% CI 0.88 to 0.97), with preterm birth RR 0.95 (95% CI 0.90 to 1.01). The 2017 Cochrane record, PMID 28407219, 723 citations, is also abstract-only in the database. The older abstract-only meta-analysis, PMID 21673856, 2011, Bulletin of WHO, 105 citations, supports the same direction.

Full-text subgroup evidence from PMID 37002655, 2023, Maternal & Child Nutrition, 14 citations, pmc, reports MMS versus IFA effects by gestational-age assessment method: overall LBW RR 0.88 (95% CI 0.86 to 0.91), preterm birth RR 0.94 (95% CI 0.88 to 1.00), and SGA RR 0.98 (95% CI 0.86 to 1.00). Effects on LBW were similar across ultrasound, prospective LMP, and recalled LMP subgroups; best-method trials showed LBW RR 0.89 (95% CI 0.85 to 0.92) and preterm RR 0.91 (95% CI 0.84 to 0.99).

Important subgroup modifier from abstract-only IPD meta-analysis PMID 29025632, 2017, Lancet Global Health, 261 citations: MMS produced greater neonatal mortality reduction for female neonates than males, RR 0.85 (95% CI 0.75 to 0.96) versus RR 1.06 (95% CI 0.95 to 1.17); benefits were larger among anemic women for LBW, RR 0.81 (95% CI 0.74 to 0.89), SGA 0.92 (95% CI 0.87 to 0.97), and 6-month mortality 0.71 (95% CI 0.60 to 0.86).

Mechanism: MMS addresses coexisting deficiencies in iron, folate, iodine, zinc, vitamins A/B/D and others that jointly constrain placental function, fetal growth, and infant nutrient endowment.

Government scaling pathway: Replace IFA in ANC procurement and counseling with UNIMMAP-like MMS where supply, regulatory approval, and budgets allow; maintain anemia screening and malaria/deworming protocols.

Key supporting papers: PMID 30873598 (2019, Cochrane, 364 cites, abstract-only); PMID 28407219 (2017, Cochrane, 723 cites, abstract-only); PMID 37002655 (2023, Maternal & Child Nutrition, 14 cites, PMC); PMID 29025632 (2017, Lancet Global Health, 261 cites, abstract-only); PMID 32075071 (2020, Nutrients, 251 cites, PMC).

5. Therapeutic zinc for childhood diarrhea

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national. Nutrition-specific, treatment.

Evidence justification: The full-text Cochrane record, PMID 27996088, 2016, Cochrane Database, 377 citations, pmc, reports that for acute diarrhea in children older than 6 months, zinc shortened duration by MD -11.46 hours (95% CI -19.72 to -3.19; 2581 children, 9 trials) and reduced diarrhea persisting to day 7, RR 0.73 (95% CI 0.61 to 0.88; 3865 children, 6 trials). In children with signs of malnutrition, duration fell by MD -26.39 hours (95% CI -36.54 to -16.23). Zinc increased vomiting in children older than 6 months, RR 1.57 (95% CI 1.32 to 1.86). Persistent diarrhea duration fell by MD -15.84 hours (95% CI -25.43 to -6.24).

Subgroup modifier: The same Cochrane review reports little evidence of benefit in children younger than 6 months: mean duration MD 5.23 hours (95% CI -4.00 to 14.45), and diarrhea at day 7 RR 1.24 (95% CI 0.99 to 1.54). It also notes that many trials were in Asian settings with high zinc-deficiency risk.

Mechanism: Zinc supports intestinal barrier repair, immune function, and electrolyte transport during diarrheal illness.

Government scaling pathway: Pair 10-14 day zinc course with ORS in IMCI/diarrhea treatment, health-facility and community-health-worker protocols, pharmacy supply, caregiver counseling.

Key supporting papers: PMID 27996088 (2016, Cochrane, 377 cites, PMC); PMID 21501441 (2011, BMC Public Health, 312 cites, PMC); PMID 36994923 (2023, Cochrane, 35 cites, abstract-only); PMID 24826920 (2014, Cochrane, 180 cites, abstract-only).

6. Breastfeeding promotion and protection

Rating: Evidence A; cost-effectiveness Very High; scalability Proven national. Nutrition-specific plus regulatory/social protection.

Evidence justification: The database includes high-authority but mostly abstract-only breastfeeding records. PMID 22895934, 2012, Cochrane, 1699 citations, abstract-only, supports exclusive breastfeeding for 6 months: no weight/length deficit signal and lower gastrointestinal infection risk. The Lancet record PMID 26869575, 2016, 7924 citations, abstract-only, reports that near-universal breastfeeding could prevent 823,000 annual child deaths and 20,000 breast-cancer deaths. A developing-country meta-analysis, PMID 35844158, 2022, Journal of Nursing Scholarship, 20 citations, abstract-only, reports breastfeeding was inversely associated with overall undernutrition, OR 0.82 (95% CI 0.68 to 0.99), and stunting, OR 0.73 (95% CI 0.55 to 0.95).

Mechanism: Breast milk supplies age-appropriate macro/micronutrients, immune factors, safe hydration, and protection from contaminated foods/water during the highest-risk period.

Government scaling pathway: Baby-Friendly Hospital Initiative, maternity protection, lactation counseling, community peer support, International Code enforcement, and integration with postnatal care.

Key supporting papers: PMID 22895934 (2012, Cochrane, 1699 cites, abstract-only); PMID 26869575 (2016, Lancet, 7924 cites, abstract-only); PMID 35844158 (2022, Journal of Nursing Scholarship, 20 cites, abstract-only); PMID 33622796 (2021, Pediatrics, 34 cites, abstract-only).

Tier 2: Strong Evidence, Scalable With Investment

7. Micronutrient powders for home fortification

Rating: Evidence A; cost-effectiveness High; scalability Proven subnational/national in many settings. Nutrition-specific.

Evidence justification: The main effect-size record is abstract-only: PMID 32107773, 2020, Cochrane, 116 citations, reports MNPs reduced anemia by 18%, RR 0.82 (95% CI 0.76 to 0.90), and iron deficiency by 53%, RR 0.47 (95% CI 0.39 to 0.56), increased hemoglobin MD 2.74 g/L (95% CI 1.95 to 3.53), and had no effect on weight-for-age, MD 0.02 (95% CI -0.03 to 0.07). Earlier abstract-only Cochrane PMID 21901727, 2011, 397 citations, reported anemia RR 0.69 (95% CI 0.60 to 0.78) and iron deficiency RR 0.49 (95% CI 0.35 to 0.67).

Implementation modifier: The full-text adherence meta-analysis PMID 34658128, 2022, Maternal & Child Nutrition, 11 citations, pmc, reports factors associated with high adherence: child age 18-36 months OR 1.45 (95% CI 1.12 to 1.88), caregiver perception that other mothers use MNPs OR 1.52 (95% CI 1.19 to 1.95), correct caregiver knowledge OR 1.36 (95% CI 1.19 to 1.57), and absence of child side effects OR 2.77 (95% CI 2.46 to 3.11).

Mechanism: Point-of-use sachets add iron and other micronutrients to semi-solid complementary foods without requiring industrial fortification.

Government scaling pathway: Target high-anemia districts through child-health days, CHWs, ECD centers or social marketing; bundle with IYCF counseling and adherence monitoring.

Key supporting papers: PMID 32107773 (2020, Cochrane, 116 cites, abstract-only); PMID 21901727 (2011, Cochrane, 397 cites, abstract-only); PMID 34658128 (2022, Maternal & Child Nutrition, 11 cites, PMC); PMID 31973225 (2020, Nutrients, 259 cites, PMC).

8. Complementary feeding counseling plus food provision where needed

Rating: Evidence A; cost-effectiveness High for counseling, Moderate-High for food provision; scalability Proven subnational. Nutrition-specific.

Evidence justification: Full-text PMID 21501443, 2011, BMC Public Health, 378 citations, pmc, reports education-only effects in food-secure populations: HAZ effect 0.21 (95% CI 0.01 to 0.41) and WAZ 0.30 (95% CI 0.05 to 0.54). Food provision with/without education in food-insecure settings improved HAZ 0.39 (95% CI 0.05 to 0.73) and WAZ 0.26 (95% CI 0.04 to 0.48). The supplementary full-text review PMID 24564534, 2013, BMC Public Health, 249 citations, pmc, reports complementary feeding provision with/without education reduced respiratory infections by 33% (95% CI 0.49 to 0.91) but did not show significant diarrhea effects.

The abstract-only update PMID 28904113, 2017, Journal of Nutrition, 127 citations, reports education/counseling improved LAZ in food-secure populations, SMD 0.11 (95% CI 0.01 to 0.22), while food supplementation with/without education improved LAZ SMD 0.08 (95% CI 0.04 to 0.13) and WLZ SMD 0.05 (95% CI 0.01 to 0.08) in food-insecure settings. The abstract-only Cochrane weaning education review PMID 32710657, 2020, reports WAZ MD 0.15 SD (95% CI 0.07 to 0.22) and HAZ MD 0.12 SD (95% CI 0.05 to 0.19) by 12 months.

Subgroup modifier: Education alone works best where household food availability is not the limiting factor; food provision is more relevant in food-insecure settings.

Mechanism: Improves meal frequency, dietary diversity, energy density, responsive feeding, and hygiene during the 6-23 month growth-faltering window.

Government scaling pathway: IYCF counseling through CHWs and primary care; targeted food/voucher provision for food-insecure households; link to growth monitoring and social protection.

Key supporting papers: PMID 21501443 (2011, BMC Public Health, 378 cites, PMC); PMID 24564534 (2013, BMC Public Health, 249 cites, PMC); PMID 28904113 (2017, Journal of Nutrition, 127 cites, abstract-only); PMID 32710657 (2020, Cochrane, 17 cites, abstract-only).

9. CMAM, RUTF/RUSF, and supplementary foods for acute malnutrition

Rating: Evidence A; cost-effectiveness High for targeted treatment; scalability Proven subnational and emergency/national programs. Nutrition-specific treatment.

Evidence justification: The database includes both SAM/MAM management and food-product comparisons. The abstract-only Cochrane review PMID 23794237, 2013, 118 citations, reports specially formulated food versus standard care increased MAM recovery, RR 1.29 (95% CI 1.20 to 1.38), reduced drop-out RR 0.30 (95% CI 0.22 to 0.39), and improved WHZ MD 0.20 (95% CI 0.03 to 0.37). LNS versus blended foods increased recovery, RR 1.10 (95% CI 1.04 to 1.16), and reduced non-recovery RR 0.53 (95% CI 0.40 to 0.69). PMID 34535798, 2021, abstract-only, reports LNS improved MAM recovery versus fortified blended foods, RR 1.05 (95% CI 1.01 to 1.09).

Full-text PMID 31906272, 2020, *Nutrients*, 109 citations, [pmc](#), synthesizes SAM/MAM management: facility-based screening/management of uncomplicated SAM had no effect on recovery and mortality; therapeutic milk F100 was comparable to RUTF for weight gain and mortality; standard RUSF had additional benefits over corn-soy blend; prophylactic antibiotics in uncomplicated SAM improved recovery and probably weight gain/mortality. Full-text PMID 24564235, 2013, *BMC Public Health*, 217 citations, [pmc](#), provides the earlier SAM/MAM review base.

Mechanism: High-energy, micronutrient-dense foods restore tissue deficits; outpatient CMAM reduces access barriers and infection risks versus prolonged inpatient care.

Government scaling pathway: MUAC screening, CHW referral, outpatient therapeutic program sites, RUTF/RUSF procurement or local production, simplified protocols, and supply-chain QA.

Key supporting papers: PMID 23794237 (2013, *Cochrane*, 118 cites, abstract-only); PMID 34535798 (2021, *Trans R Soc Trop Med Hyg*, 5 cites, abstract-only); PMID 31906272 (2020, *Nutrients*, 109 cites, *PMC*); PMID 24564235 (2013, *BMC Public Health*, 217 cites, *PMC*).

10. Small-quantity lipid-based nutrient supplements

Rating: Evidence A; cost-effectiveness Moderate-High; scalability Growing. Nutrition-specific prevention.

Evidence justification: For child prevention, the abstract-only *Cochrane* record PMID 31046132, 2019, 115 citations, reports LNS plus complementary feeding reduced moderate stunting RR 0.93 (95% CI 0.88 to 0.98), severe stunting RR 0.85 (95% CI 0.74 to 0.98), moderate wasting RR 0.82 (95% CI 0.74 to 0.91), moderate underweight RR 0.85 (95% CI 0.80 to 0.91), and anemia RR 0.79 (95% CI 0.69 to 0.90). Compared with MNP, LNS reduced anemia RR 0.38 (95% CI 0.21 to 0.68), but showed no difference for moderate stunting or wasting.

For prenatal SQ-LNS, full-text IPD meta-analysis PMID 39154665, 2024, *AJCN*, 19 citations, [pmc](#), reports SQ-LNS versus IFA/SOC increased birthweight MD 48.7 g (95% CI 26.1 to 71.2), WAZ MD 0.12 (95% CI 0.06 to 0.17), birth length MD 0.20 cm (95% CI 0.09 to 0.31), LAZ MD 0.11 (95% CI 0.06 to 0.17), head circumference MD 0.11 cm (95% CI 0.04 to 0.18), MUAC MD 0.08 cm (95% CI 0.02 to 0.14), and gestation duration MD 0.12 weeks (95% CI 0.01 to 0.24). Compared with MMS, most birth outcomes were not statistically significant; e.g., birthweight MD 27.8 g (95% CI -19.1 to 74.6).

Mechanism: Delivers energy, essential fatty acids, protein, and micronutrients in a small daily food-based dose.

Government scaling pathway: Target food-insecure districts, integrate with IYCF/ANC and supplementary feeding, ensure local production or reliable procurement, monitor sharing and adherence.

Key supporting papers: PMID 31046132 (2019, *Cochrane*, 115 cites, abstract-only); PMID 39154665 (2024, *AJCN*, 19 cites, *PMC*); PMID 30168868 (2018, *Cochrane*, 91 cites, abstract-only); PMID 31697329 (2020, *AJCN*, 76 cites, abstract-only).

11. Fortified complementary foods and fortified beverages

Rating: Evidence B+; cost-effectiveness Moderate-High; scalability Growing. Nutrition-specific/market-regulatory.

Evidence justification: Full-text PMID 35753314, 2022, *Lancet Child & Adolescent Health*, 34 citations, [pmc](#), reports fortified complementary foods reduced anemia, RR 0.57 (95% CI 0.39 to 0.82), increased hemoglobin MD 3.44 g/L (95% CI 1.33 to 5.55), and increased ferritin on log scale MD 0.43 (95% CI 0.14 to 0.72). It did not improve WAZ, MD -0.01 (95% CI -0.07 to 0.06), WLZ, MD -0.05 (95% CI -0.19 to 0.10), or LAZ, MD -0.01 (95% CI -0.21 to 0.20). Full-text PMID 26007336, 2015, *Nutrients*, 30 citations, [pmc](#), reports MMN-fortified non-dairy beverages in school-age children improved hemoglobin by 2.76 g/L (95% CI 1.19 to 4.33), ferritin by 15.42 pmol/L (95% CI 5.73 to 25.12), and reduced anemia RR 0.58 (95% CI 0.29 to 0.88), iron deficiency RR 0.34 (95% CI 0.21 to 0.55), and iron-deficiency anemia RR 0.17 (95% CI 0.06 to 0.53).

Mechanism: Adds micronutrients to foods already consumed by target age groups, especially where staple fortification may not reach infants.

Government scaling pathway: Standards for commercial complementary foods, public procurement for vulnerable groups, school feeding, and regulation to avoid displacement of breastfeeding or inappropriate marketing.

Key supporting papers: PMID 35753314 (2022, Lancet Child & Adolescent Health, 34 cites, PMC); PMID 26007336 (2015, Nutrients, 30 cites, PMC); PMID 31973225 (2020, Nutrients, 259 cites, PMC).

12. Double-fortified salt

Rating: Evidence B+; cost-effectiveness Very High; scalability Proven subnational, national potential. Nutrition-specific.

Evidence justification: The database record is abstract-only: PMID 29767699, 2018, Advances in Nutrition, 38 citations. It reports hemoglobin SMD 0.28 (95% CI 0.11 to 0.44), anemia RR 0.59 (95% CI 0.46 to 0.77), iron-deficiency anemia RR 0.37 (95% CI 0.25 to 0.54), and smaller but significant effectiveness-study hemoglobin SMD 0.03 (95% CI 0.01 to 0.05). The lower effectiveness estimate is the reason DFS is not folded into the highest LSFF rank despite an excellent cost profile.

Mechanism: Uses salt as a near-universal vehicle for iodine and iron.

Government scaling pathway: Salt iodization platform plus iron premix technology, quality control for organoleptic stability, public procurement for school meals and safety nets.

Key supporting papers: PMID 29767699 (2018, Advances in Nutrition, 38 cites, abstract-only); PMID 30880739 (2019, J Health Popul Nutr, 8 cites, abstract-only); PMID 30997493 (2019, AJCN, 215 cites, PMC).

13. Balanced protein-energy supplementation in pregnancy

Rating: Evidence B+; cost-effectiveness Moderate; scalability Requires targeting investment. Nutrition-specific.

Evidence justification: Full-text PMID 25857334, 2015, Maternal & Child Nutrition, 84 citations, pmc, reports balanced protein-energy supplementation in undernourished pregnant women improved birthweight, standardized mean difference d 0.20 (95% CI 0.03 to 0.38), with no significant benefit on birth length or head circumference and insufficient longer-term growth evidence. The full-text Cochrane overview PMID 30480324, 2018, 69 citations, reports balanced energy/protein supplementation in pregnancy may reduce stillbirth RR 0.60 (95% CI 0.39 to 0.94), increase birthweight MD 40.96 g (95% CI 4.66 to 77.26), and reduce SGA RR 0.79 (95% CI 0.69 to 0.90).

Mechanism: Addresses maternal energy and protein deficits that constrain fetal growth, particularly in chronically food-insecure populations.

Government scaling pathway: Target through ANC using food insecurity, low BMI, adolescent pregnancy, or poor gestational weight gain; link to social protection and local food procurement.

Key supporting papers: PMID 25857334 (2015, Maternal & Child Nutrition, 84 cites, PMC); PMID 30480324 (2018, Cochrane overview, 69 cites, PMC); PMID 37331760 (2023, AJCN, 21 cites, abstract-only).

14. School-age/adolescent iron, IFA, and MMN

Rating: Evidence B+; cost-effectiveness High; scalability Proven subnational. Nutrition-specific.

Evidence justification: Abstract-only PMID 24130243, 2013, CMAJ, 155 citations, reports daily iron supplementation in primary-school-aged children improved global cognitive scores, SMD 0.50 (95% CI 0.11 to 0.90), improved IQ among anemic children, MD 4.55 (95% CI 0.16 to 8.94), and reduced anemia by 50% and iron deficiency by 79%. Full-text PMID 38862227, 2024, BMJ Open, 10 citations, reports weekly IFAS improved ferritin, Hedges g 0.53 (95%

CI 0.28 to 0.78), and hemoglobin, Hedges g 0.37 (95% CI 0.01 to 0.73). Abstract-only PMID 40063075, 2025, Nutrition Reviews, 4 citations, reports iron increased hemoglobin 5.81 g/L (95% CI 4.19 to 7.44) and MMN increased hemoglobin 4.82 g/L (95% CI 2.32 to 7.32) among ages 5-24 in LMICs.

Mechanism: Corrects iron deficiency during growth and menstruation; may improve attention, school performance, and future pregnancy iron stores.

Government scaling pathway: School health programs, adolescent health days, deworming integration, menstrual-health platforms, and weekly supervised supplementation.

Key supporting papers: PMID 24130243 (2013, CMAJ, 155 cites, abstract-only); PMID 38862227 (2024, BMJ Open, 10 cites, PMC); PMID 40063075 (2025, Nutrition Reviews, 4 cites, abstract-only); PMID 31878019 (2019, Nutrients, 49 cites, PMC).

15. Preventive zinc supplementation

Rating: Evidence B+; cost-effectiveness High; scalability Proven subnational. Nutrition-specific.

Evidence justification: Full-text PMID 21501441, 2011, BMC Public Health, 312 citations, pmc , reports zinc supplementation alone had a nonsignificant all-cause mortality RR 0.91 (95% CI 0.82 to 1.01), diarrhea mortality RR 0.82 (95% CI 0.64 to 1.05), pneumonia mortality RR 0.85 (95% CI 0.65 to 1.11), diarrhea incidence RR 0.87 (95% CI 0.81 to 0.94), and pneumonia morbidity RR 0.81 (95% CI 0.73 to 0.90). Abstract-only PMID 24826920, 2014, Cochrane, 180 citations, similarly reports all-cause mortality RR 0.95 (95% CI 0.86 to 1.05), diarrhea incidence RR 0.87 (95% CI 0.85 to 0.89), vomiting RR 1.29 (95% CI 1.14 to 1.46), and no clear hemoglobin/iron benefit.

Mechanism: Supports immune function and epithelial integrity in zinc-deficient populations.

Government scaling pathway: More demanding than therapeutic zinc because it requires repeated preventive dosing or fortified delivery. Best targeted to high-deficiency areas or delivered via fortified foods/MNPs.

Key supporting papers: PMID 21501441 (2011, BMC Public Health, 312 cites, PMC); PMID 24826920 (2014, Cochrane, 180 cites, abstract-only); PMID 36994923 (2023, Cochrane, 35 cites, abstract-only); PMID 20156999 (2010, Pediatrics, 132 cites, abstract-only).

Tier 3: Promising or Platform Evidence With a Plausible Scaling Pathway

16. Financial incentives, vouchers, and cash-plus nutrition

Rating: Evidence B; cost-effectiveness Moderate; scalability Proven national where social protection exists. Nutrition-sensitive.

Evidence justification: Full-text PMID 24564520, 2013, BMC Public Health, 230 citations, pmc , found financial incentives may increase coverage of child-health interventions, including breastfeeding indicators, vaccination, care seeking, ORS use, continued feeding during diarrhea, iron, vitamin A, zinc, and growth monitoring, but evidence quality was low. It found more pronounced effects where user fees were removed and where incentives were tied to health education or visits. Full-text PMID 31666032, 2019, BMC Public Health, 25 citations, pmc , reports 17 stunting/economic-development studies; 76% reduced stunting and 65% reported both stunting and economic successes, but designs were mixed.

Mechanism: Relaxes income and access constraints, increases service uptake, and can improve food purchases when combined with counseling.

Government scaling pathway: Use existing social registries and payment systems; make “cash plus” packages include IYCF, ANC, growth monitoring, and referral rather than relying on cash alone.

Key supporting papers: PMID 24564520 (2013, BMC Public Health, 230 cites, PMC); PMID 31666032 (2019, BMC Public Health, 25 cites, PMC).

17. WASH integrated with nutrition

Rating: Evidence B; cost-effectiveness Moderate; scalability Requires infrastructure investment. Nutrition-sensitive.

Evidence justification: Full-text PMID 31272479, 2019, Italian Journal of Pediatrics, 54 citations, pmc , reports WASH effects on child linear growth vary by age and intervention type. Abstract-only Cochrane PMID 23904195, 2013, 377 citations, is a high-citation source but only abstract is available in the database. Mechanistic text in the full-text review links diarrhea, parasitic infection, and environmental enteric dysfunction to poor nutrient absorption and growth.

Subgroup modifier: Single household-level WASH components have smaller and less consistent nutrition effects than infrastructure-level changes or combined packages, and large recent trials outside this database have made direct stunting effects more uncertain. In this database, WASH is best treated as enabling infrastructure rather than a standalone nutrition cure.

Mechanism: Reduces diarrheal disease, helminth exposure, and environmental enteric dysfunction.

Government scaling pathway: Align nutrition hot-spot mapping with water/sanitation investment, handwashing promotion, safe child feces disposal, and health/nutrition visits.

Key supporting papers: PMID 31272479 (2019, Italian Journal of Pediatrics, 54 cites, PMC); PMID 23904195 (2013, Cochrane, 377 cites, abstract-only).

18. Nutrition-sensitive agriculture, eggs, and food systems

Rating: Evidence B-/C+; cost-effectiveness Moderate; scalability Growing. Nutrition-sensitive with nutrition-specific food supplementation overlap.

Evidence justification: Full-text agriculture review PMID 22251864, 2012, BMJ, 396 citations, pmc , found little evidence of positive effects on stunting, wasting, and underweight, though 19 of 23 studies improved diet composition and pooled serum retinol increased by 2.4 ug/dL. Full-text egg meta-analysis PMID 42137196, 2026, Journal of Nutritional Science, 0 citations, pmc , reports egg-based intervention improved WAZ MD 0.33 (95% CI 0.11 to 0.55) and WHZ MD 0.30 (95% CI 0.12 to 0.48), but not HAZ MD 0.05 (95% CI -0.05 to 0.14). The small and recent evidence base keeps this out of Tier 2.

Mechanism: Improves household availability/access to animal-source foods, fruits/vegetables, and micronutrient-rich foods.

Government scaling pathway: Agriculture extension plus behavior-change communication, school meals, poultry/egg value chains, biofortification, and safety-net procurement.

Key supporting papers: PMID 22251864 (2012, BMJ, 396 cites, PMC); PMID 42137196 (2026, Journal of Nutritional Science, 0 cites, PMC); PMID 32270495 (2020, Cochrane, 22 cites, abstract-only).

19. Growth monitoring and promotion

Rating: Evidence C+; cost-effectiveness Moderate as a platform, low/uncertain alone; scalability Proven national. Nutrition-specific platform.

Evidence justification: Full-text Cochrane PMID 37823471, 2023, 28 citations, pmc , found evidence very uncertain. For GMP with supplementary feeding, one summary table reports WAZ MD -0.07 (95% CI -0.19 to 0.06) and HAZ MD -0.15 (95% CI -0.34 to 0.04) at 12 months, indicating little or no difference. Other tabled outcomes include normal nutrition status RR 1.11 (95% CI 0.95 to 1.30), first-degree malnutrition RR 0.86 (95% CI 0.71 to 1.05), and second-degree malnutrition RR 1.14 (95% CI 0.59 to 2.19). GMP should therefore be valued as a platform for counseling, supplementation, and referral rather than as a standalone growth intervention.

Mechanism: Detects faltering and creates a counseling/referral moment; measurement alone does not treat undernutrition.

Government scaling pathway: Redesign national GMP to include decision support, referral pathways, commodity links, and quality counseling.

Key supporting papers: PMID 37823471 (2023, Cochrane, 28 cites, PMC); PMID 24564520 (2013, BMC Public Health, 230 cites, PMC).

20. Nutritional management of low-birthweight/preterm infants

Rating: Evidence C+; cost-effectiveness Moderate; scalability Growing. Nutrition-specific neonatal care.

Evidence justification: Full-text PMID 39591949, 2025, Neonatology, 11 citations, pmc , synthesizes LBW/preterm feeding strategies in low and lower-middle income countries. The extracted full text includes a no-difference result for time to regain birthweight overall, MD -1.48 days (95% CI -4.73 to 1.77; 74 infants, 3 studies), and in LMIC settings, MD -4.5 days (95% CI -9.0 to 0.0; truncated in extracted text). Evidence is important but still heterogeneous.

Mechanism: Early and adequate breastmilk feeding, kangaroo mother care support, fortification where appropriate, and management of small/nutritionally at-risk infants.

Government scaling pathway: Facility newborn units, kangaroo mother care wards, postnatal follow-up, lactation support, and referral for small infants under 6 months.

Key supporting papers: PMID 39591949 (2025, Neonatology, 11 cites, PMC); PMID 34407128 (2021, PLoS One, 65 cites, PMC).

21. Nutritional support for adults living with HIV

Rating: Evidence C+; cost-effectiveness Unknown/Moderate; scalability Requires investment. Nutrition-specific clinical support.

Evidence justification: Full-text PMID 40460180, 2025, PLoS One, 3 citations, pmc , reports nutritional interventions for adults living with HIV improved BMI SMD 0.42 (95% CI 0.03 to 0.81), fat mass SMD 0.21 (95% CI 0.07 to 0.34), fat-free mass SMD 0.33 (95% CI 0.19 to 0.46), and CD4 SMD 0.54 (95% CI 0.01 to 1.07), with no effect on weight, viral load, or ART adherence. This is relevant to clinical nutrition rather than child stunting policy.

Mechanism: Improves energy/protein intake and body composition in wasting or food-insecure PLWHA.

Government scaling pathway: Targeted nutrition assessment, counseling, and support within ART programs, with food assistance reserved for undernourished or food-insecure patients.

Key supporting papers: PMID 40460180 (2025, PLoS One, 3 cites, PMC); PMID 30480324 (2018, Cochrane overview, 69 cites, PMC).

22. Aflatoxin-reduction agricultural and nutrition education

Rating: Evidence C; cost-effectiveness Unknown; scalability Requires investment. Nutrition-sensitive.

Evidence justification: Abstract-only Cochrane PMID 32270495, 2020, 22 citations, reports that agricultural education practices such as sorting, de-hulling, drying sheets, and insecticides may improve infant WAZ by MD 0.57 (95% CI 0.16 to 0.98; 1 study, 249 participants) and may reduce underweight by 6.7 percentage points (95% CI -12.6 to -1.4), but certainty was very low.

Mechanism: Reduces aflatoxin exposure that may impair growth and immune function.

Government scaling pathway: Agricultural extension, storage/drying technology, food safety surveillance, and integration with nutrition education in high-aflatoxin crop systems.

Key supporting papers: PMID 32270495 (2020, Cochrane, 22 cites, abstract-only).

Cross-Cutting Findings

1. The strongest actionable evidence is concentrated in the first 1000 days. Pregnancy supplementation, breastfeeding, complementary feeding, MNP/LNS, and acute malnutrition treatment dominate the high-ranking intervention set.
2. Delivery platforms explain scalability. The top-ranked interventions use existing platforms: ANC for IFA/MMS, immunization or child health days for VAS, food regulation for fortification, IMCI/primary care for zinc, and maternity/community platforms for breastfeeding.
3. Adherence and coverage are often bigger problems than efficacy. IFA, MNPs, complementary feeding, and breastfeeding all show that counseling, side-effect management, social norms, and reliable supplies determine real-world impact.
4. Food-based interventions improve growth when food is the constraint, but effects are more context-specific. Complementary food provision, LNS, balanced energy-protein, and eggs show better effects in food-insecure or undernourished groups than in general populations.
5. Nutrition-sensitive interventions are plausible but less direct. Cash, agriculture, and WASH address underlying causes, but the database supports them most strongly as “plus” platforms paired with nutrition-specific services, not as substitutes for supplementation, fortification, IYCF, and treatment.
6. Some long-standing national platforms need redesign rather than expansion. Growth monitoring and IMCI are scalable, but the evidence suggests limited nutrition impact unless measurement is tied to counseling, commodities, referral, and follow-up.

Classification Notes

- Primary-tier classification is mostly validated by PubMed publication types: Meta-Analysis and Systematic Review dominate the top-ranked records.
- MeSH terms support nutrition-specific classification for micronutrient supplementation, dietary supplements, pregnancy, child nutrition disorders, zinc, vitamin A, anemia, and developing countries.
- Full-text effect sizes are available for core intervention papers including LSFF, pregnancy education/IFA adherence, SQ-LNS, VAS, complementary feeding, WASH, growth monitoring, HIV nutrition, egg supplementation, adolescent WIFAS, and acute malnutrition management.
- Key abstract-only evidence was unavoidable for several high-authority Cochrane/Lancet records: breastfeeding, newer VAS Cochrane updates, MNP Cochrane updates, maternal MMS Cochrane reviews, DFS, child LNS, and several school-age/adolescent supplementation reviews.
- The provided top 100 did not include a separate CEA tier despite the prompt description. Cost-effectiveness ratings should therefore be interpreted as program-priority ratings, not extracted ICER comparisons.